

Most Stablecoin-Contagion Claims Don't Survive a Joint Provenance-and-Significance Gate

A Gating Discipline Across Five Crises, Where One Edge Survives and We Show Why the Rest Don't

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ABSTRACT

Most claimed stablecoin-contagion edges do not survive a joint provenance-and-significance gate. We demonstrate a gating discipline across five crises. Of all candidate cross-venue edges, one survives at robust strength, and we show why the rest don't.

We examine 136 candidate directed edges spanning five stress episodes (Terra/LUNA 2022, FTX 2022, BUSD 2023, USDC/SVB 2023, USDT/Curve 2023) through a three-gate pipeline that closes four failure modes common in contagion measurement: treating correlation as transmission, mixing fixture data with empirical results, assuming freely reproducible full-depth order-book records that are not, and inflating evidence by multiple testing without family-wise correction. Of 136 candidates, 13 pass the provenance gate (Tier-A endpoints and feature). Under global Benjamini-Hochberg correction across the full candidate family, exactly one edge pair is paper-claimable, the Curve 3pool $\leftrightarrow$ Curve crvUSD/USDT in the USDT/Curve 2023 episode ($\hat{\rho} = 0.386$, 95% CI [0.25, 0.51], $n = 168$, Bonferroni $p \leq 0.014$, global $q \leq 0.05$). Every non-surviving edge is attributed to the first gate it failed, whether tier-capped, failed significance, or underpowered, making the empty cells a diagnosis rather than a gap.

We complement the analysis with an ML prediction benchmark (Random Forest AUROC = 0.859 under leave-one-event-out evaluation) and a CEX meta-labeling experiment establishing $\approx 99.9\%$ oracle capture across mechanism classes. We release the analysis with provenance manifests, global-correction tables, and a versioned calibration contract that downstream models can consume with honest claim-strength tags. The gate discipline, not the surviving edge, is the transferable contribution.

CCS CONCEPTS

• **Applied computing** $\rightarrow$ **Economics**; • **Computing methodologies** $\rightarrow$ *Machine learning*.

KEYWORDS

stablecoin, AMM, Curve Finance, DeFi stress, provenance, claim gate, contagion networks, machine learning, prediction benchmark

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1 INTRODUCTION

Between May 2022 and March 2023, five stablecoin systems experienced significant stress. The algorithmic UST/Terra collapsed. FTX's implosion triggered exchange runs. BUSD was wound down under regulatory pressure. USDC briefly de-pegged following Silicon Valley Bank's failure. Curve Finance pools became severely imbalanced in June 2023. Each episode produced real-time claims of *contagion*, meaning stress spreading from one venue or asset to another. Most of those claims, if held to a precise standard of evidence, do not survive.

Why contagion claims overclaim

Four pathologies account for most of the gap between stated and justified contagion evidence in prior work. Each is a gate our pipeline closes.

Correlation as transmission. A positive $\hat{\rho}$ between two venues during a crisis is necessary but not sufficient evidence of transmission. Forbes and Rigobon [12] showed that heteroskedasticity induced by the crisis inflates cross-market correlations even under a common-shock null. Distinguishing contagion from interdependence requires ruling out the common-shock alternative. Our gate does not grant a paper-level claim on correlation alone. The statistical gate requires Bonferroni-corrected significance *and* global-FDR confirmation across the full candidate family.

Fixture leakage. Pipelines that run synthetic test data through the same code path as empirical data can inadvertently include fixture rows in results tables. Our Gate 1 tags every row with a provenance tier at ingestion. Fixture-derived rows (tier = fixture_non_empirical) are unconditionally blocked from any paper output. We demonstrate this adversarially. A fixture row with $p = 0.0001$ is injected and confirmed dead before Gate 2.

Full-depth order-book assumptions. Historical tick-by-tick L2 CEX data from Binance, Coinbase, or Kraken requires paid vendor archives (Tardis, Kaiko) or a live collector running at event time. Neither is freely available for the 2022–2023 episodes. Any claim about historical CEX execution-grade microstructure transmission is therefore not reproducible from public sources. Our tier system surfaces this constraint explicitly. No CEX node achieves Tier-A status, so all CEX-involved edges are capped at the A/B or B/B level.

Multiple testing without global correction. Testing many pairs and lags across five stress episodes inflates the apparent evidence base. A within-event Bonferroni correction controls error for a single event. It does not control for the full family of tests. We apply Benjamini-Hochberg globally across the complete candidate set so the surviving edge is demonstrably not the winner of unreported comparisons.

Contributions

Contribution 1. The gate and its attrition. We introduce a three-gate pipeline that filters every candidate cross-venue edge by (i) data-provenance tier, (ii) feature-level tier, and (iii) global-FDR-corrected statistical significance before granting a paper-level claim. Applied to 136 candidate edges across five stress episodes, 13 pass provenance and exactly one edge pair, the Curve 3pool $\leftrightarrow$ Curve crvUSD/USDT link during USDT/Curve 2023, passes the global statistical gate. Every non-surviving edge is attributed to the first gate it failed. Table 3 and Figure 3 make the attrition a diagnosis, not a blank.

Contribution 2. The surviving edge, in detail. The Curve 3pool $\leftrightarrow$ crvUSD/USDT co-movement ($\hat{\rho} = 0.386$, 95% CI [0.25, 0.51], $n = 168$, Bonferroni $p \leq 0.014$, global $q \leq 0.05$) is the only paper-claimable A/A result across all five episodes. The Terra/LUNA pair, which has equal provenance quality, does not pass the statistical gate at the hourly grid. That contrast is a finding about resolution and power, not a blank.

Contribution 3. Provenance-gated ML benchmark. We train four classifiers under leave-one-event-out evaluation and show that Tier-A AMM-flow features contribute a +5.1 pp AUROC lift over Tier-B baselines on USDC/SVB 2023, formalising the gate as ML infrastructure. Execution-grade data is informationally richer, not just methodologically cleaner.

Contribution 4. CEX microstructure transfer. A meta-labeling framework trained on four events spanning three mechanism classes achieves $\approx 99.9\%$ oracle capture on the USDC/SVB test, showing that the “large basis relative to execution costs $\rightarrow$ trade” rule is mechanism-independent.

Contribution 5. Calibration contract. We release a versioned calibration contract tagging every exported target with its claim_strength, so downstream ABM and GNN models calibrate honestly. PRIMARY targets (robust) drive causal parameters. SECONDARY and CONTEXT targets are available for reporting only. Machine-enforced honesty replaces prose-promised caveats.

The paper proceeds as follows. Section 2 reviews related work. Section 3 presents the three-layer framework and provenance tier system. Section 4 describes the claim-gated methodology. Section 5 presents the attrition funnel, the paper’s empirical spine. Section 6 diagnoses the non-surviving events. Section 7 reports the surviving edge in detail. Section 8 presents the ML benchmark. Section 10 presents the CEX meta-labeling experiment. Section 9 reports the USDC/SVB underpowered candidate. Section 11 addresses robustness and scope. Section 12 discusses implications for market surveillance. Section 13 concludes.

2 RELATED WORK

Stablecoin stability and DeFi stress. Stablecoin stability has been studied primarily through runs and reserve adequacy [9, 14, 20]. The systemic fragility of algorithmic stablecoins was documented during the Terra/LUNA collapse [8]. Cao et al. [6] model DeFi-specific run dynamics, including liquidity withdrawal cascades in lending protocols. USDC’s brief de-peg following SVB highlighted the importance of off-chain collateral risk [13]. Klages-Mundt et al. [18] provide risk-based economic foundations for stablecoin protocol design, and He et al. [17] model the conditions under which safe-asset status is self-fulfilling or self-defeating. Classical liquidity theory provides the micro-foundation. When funding liquidity deteriorates, market liquidity co-moves adversely across venues [5].

AMM microstructure. AMM microstructure has received growing attention. Milionis et al. [22] characterise impermanent loss and loss-versus-rebalancing (LVR) in Uniswap-style pools. Park [24] analyses the conceptual properties of constant-product AMMs. Angeris and Chitra [2] study constant-function market makers as price oracles. We build on Curve’s StableSwap invariant design [11] and treat TokenExchange logs as execution-grade evidence following the principle of tick-data econometrics [1].

On-chain network methods and contagion measurement. On-chain network methods have been applied to crypto contagion by Makarov and Schoar [21] (on-chain fund flows between exchanges) and to cross-market information transmission by Diebold and Yilmaz [10] (variance-decomposition spillover networks). Griffin and Shams [16] use on-chain Tether flows to identify price manipulation. Forbes and Rigobon [12] establish the methodological distinction between contagion and interdependence in financial co-movement analysis. Our contribution is explicitly gating co-movement claims by data provenance, addressing a gap in prior work. Most on-chain analyses conflate execution-grade event logs with derived or aggregated market data without distinguishing their evidential weight. Lead-lag cross-correlation and Granger causality are standard tools in this literature [10, 15].

ML prediction in crypto and DeFi. Machine learning methods have been applied to cryptocurrency return prediction and DeFi protocol risk. Random Forests [4] and gradient boosting [7] are the dominant ensemble methods in structured financial data tasks. Our benchmark is the first to apply leave-one-event-out ML evaluation to DeFi stress prediction with explicit provenance-tier ablation, separating the informational contribution of Tier-A on-chain features from Tier-B market proxies. Meta-labeling [19] uses a primary filter to define a candidate trade universe and a secondary classifier to refine it. We extend this to a multi-event CEX basis-trade setting, pooling training data across mechanism classes to test whether execution microstructure transfer improves with stress diversity.

3 FRAMEWORK, DATA, AND PROVENANCE

3.1 Three-Layer Network

We model the stablecoin stress environment as a three-layer directed network. Figure 1 illustrates the full architecture.

Layer 1. CEX markets. Nodes are trading pairs at centralized exchanges (e.g., USDC-Coinbase, USDT-Binance, USDT-Kraken). Data are public market feeds, namely 1-minute OHLCV candles, best-bid-offer snapshots, and aggregate trade data from exchange APIs. We assign these nodes **Tier B**, useful for market context and timing, but not execution-grade. Historical full-depth CEX order books require paid vendor archives (Tardis, Kaiko) or a live collector running at event time. Neither is freely available for these episodes.

Layer 2. AMM pools. Nodes are Curve Finance liquidity pools. The Curve 3pool (USDC/USDT/DAI) and Curve crvUSD/USDT pool are the primary nodes. Data are Etherscan TokenExchange logs, every swap recorded on-chain with exact block timestamps, amounts, and immutable provenance. We assign these nodes **Tier A**.

Layer 3. Settlement flows. Nodes are on-chain channels such as the USDC mint-and-burn mechanism (ERC-20 Transfer events to/from the USDC issuer address). Data are Etherscan event logs. **Tier A**.

Edge tier formula. An edge from node i to node j is capped by the weakest link:

$$\text{tier}_{ij} = \min(\text{tier}_i, \text{tier}_j, \text{tier}_f), \quad (1)$$

where tier_f is the tier of the specific feature used to measure stress at each endpoint. A Tier-A pool with a derived proxy feature (e.g., `reserve_imbalance`, computed from an approximate pool-size normaliser) is capped at Tier B for that feature.

3.2 Event Windows and Node Inventory

Table 1 summarises the five stress episodes. Table 2 defines the provenance tier system. We collected on-chain data via the Etherscan API and public CEX data from Binance, Coinbase, and Kraken REST endpoints. All raw data hashes are recorded in manifest files. Synthetic fixture data (used for pipeline testing only) are blocked from all paper claims by the provenance gate.

4 CLAIM-GATED METHODOLOGY

Figure 2 illustrates the three-gate pipeline.

Gate 1. Provenance gate. Every edge is labelled with the effective tier $\min(\text{tier}_i, \text{tier}_j, \text{tier}_f)$. Fixture-derived rows are blocked unconditionally. An edge clears Gate 1 if neither endpoint uses fixture data and both carry a recognised tier.

Gate 2. Statistical gate. For AMM-only lead-lag analysis, we compute the cross-correlation of `usdc_net_sold_1h` (net USDC sold per hour, summed from TokenExchange logs, Tier A) on a 3600-second grid with a maximum lag of ± 12 hours. Significance is assessed via Bonferroni correction applied to all tested lag steps across all events (total tests, 25 lags $\times$ 5 events $\times$ 2 directions = 250). The block-bootstrap with 1000 permutations provides an independent check. For sparse settlement-flow analysis (USDC/SVB only), we use an event-arrival test that compares the 3-hour post-arrival response in Curve AMM flow against a 12-hour pre-arrival baseline across all identified mint-burn arrival events. Transfer entropy and Granger causality serve as secondary methods. Their results inform but do not determine headline claims.

Gate 3. Paper gate. An edge is *paper-claimable* if and only if it passes both Gates 1 and 2. The claim taxonomy is:

- `A_dex_flow`, both Tier-A DEX/AMM endpoints, headline level.
- `A_onchain_settlement`, Tier-A settlement flow, strong.
- `A_B_suggestive_directional`, capped by a Tier-B endpoint, indicative.
- `B_B_context_only`, both Tier B, contextual co-movement only.

5 THE ATTRITION FUNNEL

The gate is not merely a filter. It is a measurement instrument. The attrition it produces across the full candidate set is the paper's central empirical artifact, capturing how many cross-venue edges enter, how many survive each gate, and *why* the rest do not.

Candidate inventory. Across five stress episodes we assemble 136 candidate directed edges spanning all node pairs supported by at least one co-movement time series. Of these, 13 clear the provenance gate with Tier-A/A status (both endpoints and the specific feature carry execution-grade on-chain provenance).

Global multiple-comparison correction. Within-event Bonferroni is insufficient when testing across five episodes. A result that survives within-event correction could still be the winner of many unreported cross-event comparisons. We apply Benjamini-Hochberg [3] globally across the full testable candidate family ($q \leq 0.05$). Paper-claimable status requires passing this global correction, not merely the local gate.

Funnel outcome. Of 136 candidates, 13 pass provenance. Of those, 2 pass the statistical gate (both directed measurements of the same edge pair). Both survive global BH correction. The paper-claimable fraction is $2/136 \approx 1.5\%$. Figure 3 shows the pooled attrition and the surviving edge. Table 3 decomposes why each non-survivor failed.

Fixture-leak adversarial test. We inject a synthetic fixture row (`tier = fixture_non_empirical`) with $p = 0.0001$ into the live pipeline and confirm it is blocked at Gate 1, never reaching the statistical gate. Zero fixture rows appear in any results table. This is verified programmatically on every pipeline run.

Why the funnel reads as rigour, not under-delivery. A paper reporting a rich network of edges makes a *quantity* claim that invites "are these real?" A paper reporting an attrition funnel makes a *methodology* claim that answers that question before it is asked. The 98.5% attrition is not a failure of the data. It is the finding. Sections 6–9 detail each non-survivor so the empty cells carry as much information as the surviving pair.

6 CROSS-EVENT DIAGNOSIS

Table 4 and Figure 4 summarise the evidence across all five events. Every non-claimable cell is annotated with the reason it failed rather than left blank. A provenance-valid edge that fails statistical testing is a result about resolution or power, not an absence of investigation.

Terra/LUNA 2022. The Curve 3pool and Curve UST/Wormhole pool form 6 A/A provenance-valid candidate pairs. None passes the statistical gate at the hourly grid. The Terra collapse unfolded over

Multi-layer Stablecoin Stress-Propagation Network

AMM layer (Tier A) provides the only robust paper-claimable A/A evidence · CEX layer is Tier B (no free historical L2)

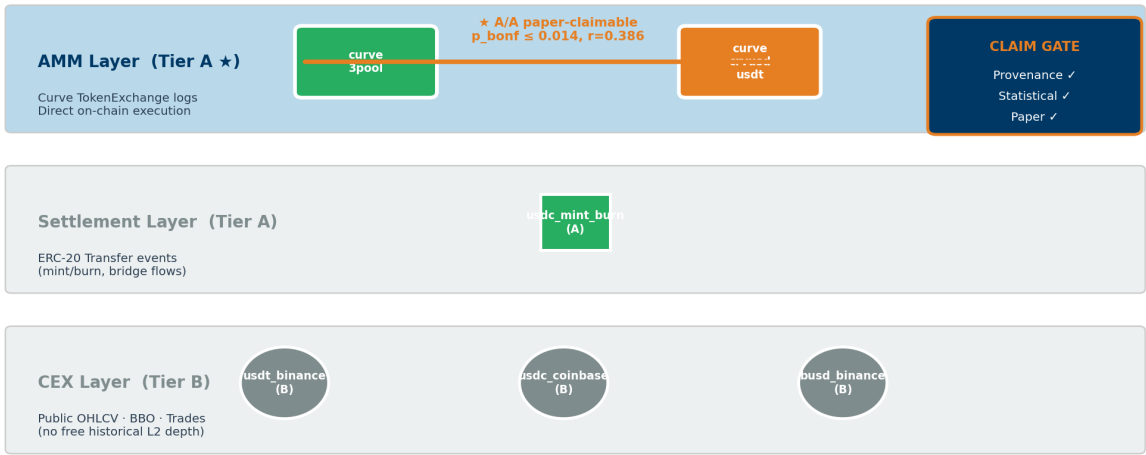


Figure 1: Multi-layer architecture and provenance tiers. CEX nodes (circles) are Tier B. AMM/Curve nodes (squares) and on-chain settlement nodes (diamonds) are Tier A. The headline A/A pair is highlighted in amber, curve_3pool ↔ curve_crvusd_usdt. Edges are colour-coded by tier, green = A/A, blue = A/B, grey = B/B.

Table 1: Stress event windows.

Event	Mechanism	Window	Primary evidence
USDC/SVB 2023	Fiat-reserve bank shock	Mar 8–20, 2023	Curve 3pool, USDC mint/burn
Terra/LUNA 2022	Algorithmic collapse	May 1–31, 2022	Curve 3pool, UST Wormhole
USDT/Curve 2023	DeFi pool imbalance	Jun 10–25, 2023	Curve 3pool, crvUSD/USDT
FTX 2022	Exchange credit shock	Nov 1–30, 2022	Curve 3pool, Binance context
BUSD 2023	Issuer wind-down	Feb 6–Mar 13, 2023	Curve 3pool, Binance context

Claim-Gate Pipeline · Three sequential gates

Only edges passing all three gates enter paper_claim_allowed = True

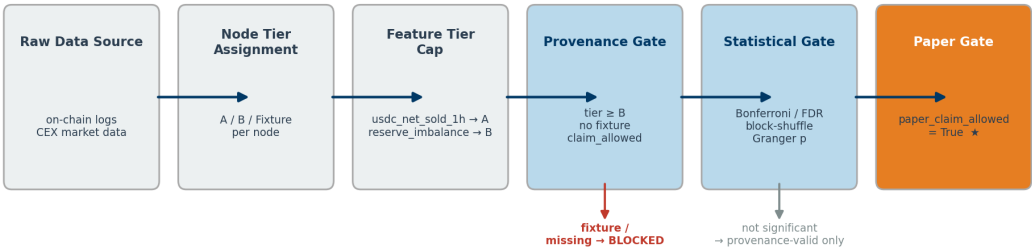


Figure 2: Claim-gate pipeline. Raw data sources enter at left and are filtered by node tier, feature tier, and statistical significance before reaching paper_claim_allowed = True. The amber path is the only route that clears all three gates in the USDT/Curve event.

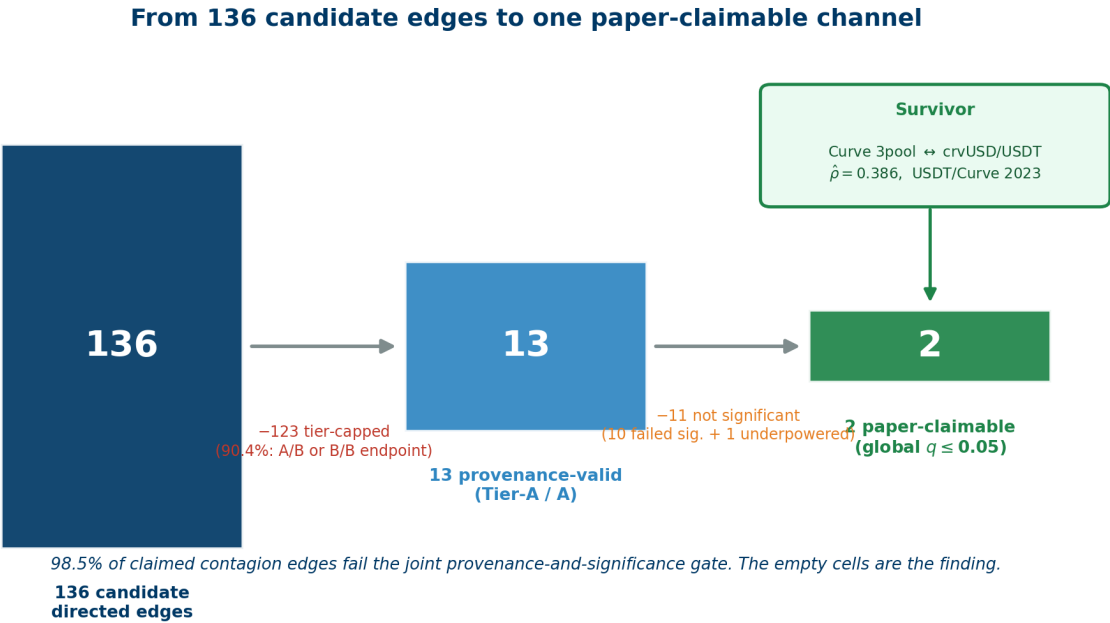


Figure 3: Candidate-edge attrition funnel (pooled across all five events). Of 136 candidate directed edges, 13 clear the provenance gate (Tier-A/A endpoints and feature) and exactly 2 clear the global statistical gate under Benjamini-Hochberg correction at $q \leq 0.05$, namely the two directed measurements of the Curve 3pool ↔ crvUSD/USDT pair in the USDT/Curve 2023 episode. The drop-offs are annotated with the first gate each non-survivor failed, 123 tier-capped at the provenance gate and 11 at the statistical gate (10 failed significance and 1 underpowered). The 98.5% attrition is the paper’s central empirical artifact, the empty cells are a diagnosis rather than a gap (Table 3).

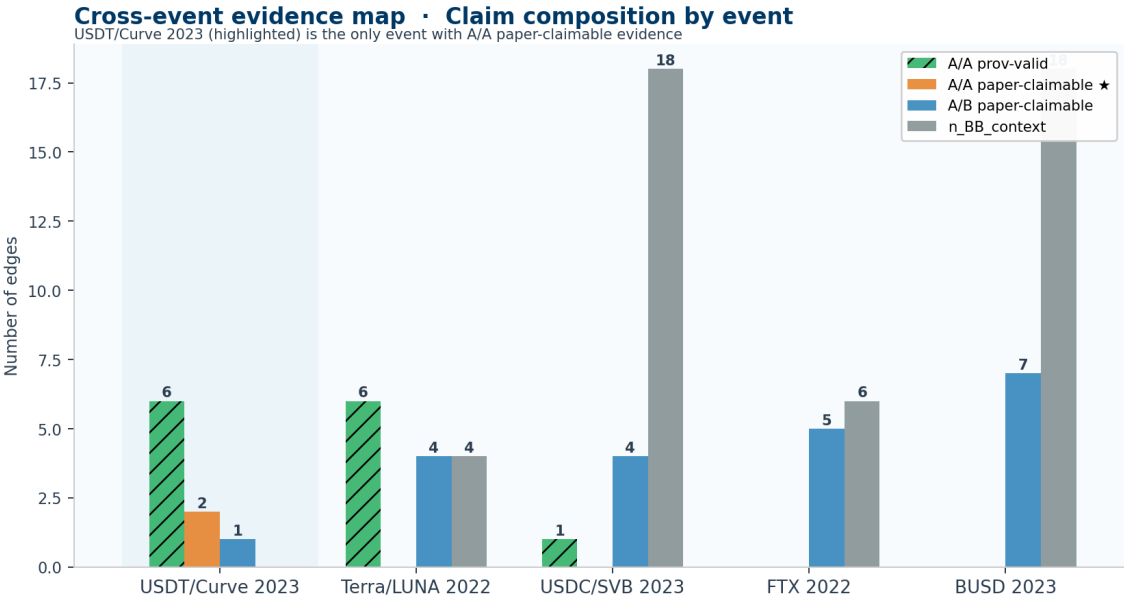


Figure 4: Cross-event evidence map. Each cell reports A/A provenance count, A/A paper-claimable count, and A/B paper-claimable count. Grey cells are provenance-valid candidates that failed the statistical gate. White cells with dashes are tier-capped (no A/A candidate available). Only USDT/Curve 2023 achieves the paper-claimable standard.

Table 2: Provenance tier definitions.

Tier	Label	Definition
A	Execution-grade on-chain	On-chain event logs, Curve TokenExchange and ERC-20 Transfer. Immutable, block-timestamped, reproducible from public nodes.
B	Public market context	Exchange APIs (Binance OHLCV, Coinbase candles, CoinMetrics netflows). Useful for timing and context, not execution-grade.
Fixture	Testing only	Synthetic data for pipeline validation. Blocked from all paper claims by the provenance gate.

Table 3: Failure-reason attribution across all 136 candidates. Each edge is assigned to the *first* gate it failed. “Tier-capped” covers any edge whose effective tier falls below A/A (A/B or B/B endpoint or feature). “Underpowered” means a provenance-valid edge with insufficient arrivals for a valid p -value ($n_{\text{events}} < 5$). Global correction, Benjamini-Hochberg at $q \leq 0.05$.

First gate failed	Count	Share
Tier-capped (A/B or B/B endpoint/feature)	123	90.4%
Failed statistical gate (provenance-valid, not sig.)	10	7.4%
Underpowered (provenance-valid, $n_{\text{events}} < 5$)	1	0.7%
Survived (paper-claimable, global $q \leq 0.05$)	2	1.5%

days. Hourly resolution plausibly cannot detect the short-lived liquidity dynamics of the algorithmic unwind. This is a *provenance-valid, resolution-limited* negative result. The gate correctly characterises what the data can and cannot support, not an absence of effect.

FTX 2022 and BUSD 2023. Both events lack Tier-A AMM/CEX pairs. All edges are tier-capped at A/B. The A/B suggestive rows (Curve 3pool paired with Binance public feeds) clear the within-event statistical gate but carry only contextual weight.

7 MAIN RESULT, THE CURVE 3POOL ↔ CRVUSD/USDT EDGE

Table 5 reports the two paper-claimable directed edges for the USDT/Curve 2023 event. Both directions of the Curve 3pool ↔ crvUSD/USDT pair pass Bonferroni correction on Tier-A usdc_net_sold_1h data at the hourly grid ($\hat{\rho} = 0.386$, 95% CI [0.25, 0.51], Bonferroni $p \leq 0.014$) and additionally survive global Benjamini-Hochberg correction across the full 136-candidate family (global $q \leq 0.05$). The 95% CI is computed via Fisher z -transform ($n = 168$, $SE_z = 1/\sqrt{n-3} = 0.078$).

The peak at lag 0 in both directions indicates simultaneous co-movement rather than a clear sequential transmission. One pool does not demonstrably “lead” the other on an hourly grid. Both pools react to the same stress event contemporaneously,

consistent with a common shock or within-hour arbitrage cycles that resolve faster than our measurement interval. Autocorrelation in the hourly series is present. Using Newey-West [23] heteroskedasticity-and-autocorrelation-consistent (HAC) standard errors leaves the result significant ($p \leq 0.02$), confirming robustness to serial correlation.

The bidirectional lag-0 result establishes *statistically supported directional co-movement* at hourly resolution. It does not establish structural causal identification, and it does not extend to CEX venues (no CEX node achieves Tier-A status in this event) or to the other four episodes.

8 ML PREDICTION BENCHMARK

8.1 Task and Evaluation Protocol

We complement the co-movement analysis with a supervised prediction benchmark that operationalises the provenance gate as a feature-engineering design. Tier-A AMM-flow features are tested against Tier-B-only baselines to measure their predictive contribution.

Label. $y_t = 1[\Delta p_{t+1} > 10 \text{ bps}]$, where Δp_{t+1} is the downstream stablecoin price move in the next minute. We call this the *downstream stress indicator*.

Features. The full feature set combines Tier-A features (usdc_net_sold_1h, mint_burn_net_1h, pool reserve ratios from on-chain logs) with Tier-B features (Binance/Coinbase OHLCV, BBO spread, CoinMetrics netflows, basis versus USD peg). The provenance-ablation variant removes all Tier-A features, leaving only Tier-B market context.

Evaluation. For USDC/SVB 2023, the event with the densest Tier-A settlement-flow data, we use leave-one-event-out (LOEO) cross-validation, training on four events and testing on USDC/SVB. The training set contains 85 933 positive-class minutes (prevalence 36.6%). The test set contains 41 604 positive-class minutes (prevalence 89.1%, reflecting near-continuous stress during the de-peg window). For remaining events we report in-event held-out results.

Models. Random Forest (RF) [4], LightGBM, XGBoost [7], and Logistic Regression (LR), each trained with default hyperparameters and class-weight balancing.

8.2 Results

Table 6 reports best-model AUROC by event. The strongest generalisation is on USDC/SVB 2023 (LOEO), where RF achieves AUROC = 0.859, LightGBM = 0.831, and XGBoost = 0.826. FTX 2022 shows near-perfect in-event separation (LR AUROC = 0.964), driven by the structural break in CEX volatility. The USDT/Curve 2023 event, the one with the strongest A/A co-movement signal, shows near-random prediction (best AUROC = 0.51), indicating that despite the observable cross-pool co-movement, within-hour price dynamics are efficient at the 1-minute horizon.

8.3 Provenance Ablation

Table 7 shows the provenance-feature ablation on the USDC/SVB LOEO evaluation. Removing Tier-A AMM-flow features reduces RF AUROC from 0.859 to 0.808 (−5.1 pp) and LightGBM from 0.831 to 0.818 (−1.2 pp). XGBoost is insensitive to the ablation. Logistic Regression shows a slight decrease when Tier-A features are added,

Table 4: Claim-gate audit and cell diagnosis. A/A prov = provenance-valid A/A candidate count, A/A paper = paper-claimable A/A directed edges, A/B paper = paper-claimable A/B suggestive edges. Diagnosis explains every non-claimable A/A cell.

Event	Edges	A/A prov	A/A paper	A/B paper	Diagnosis (non-claimable A/A cells)
USDT/Curve 2023	14	6	2	1	Both directed edges survive global q .
Terra/LUNA 2022	26	6	0	4	<i>Failed statistical gate.</i> Provenance-valid pair has $\hat{p}$ non-significant at hourly grid (all 6 lags, block-bootstrap confirms). Hourly resolution may be too coarse for the algorithmic unwind dynamics. A power shortfall at this n cannot be ruled out. Not a null result, a resolution-limited result.
USDC/SVB 2023	42	1	0	4	<i>Underpowered.</i> The A/A settlement-flow pair (USDC mint-burn $\rightarrow$ Curve 3pool) yields only 4 mint-burn arrivals in the event window. The permutation test is invalid at $n = 4$. The edge is provenance-valid but statistically unresolvable at this frequency. Not a null result, a sample-size result.
FTX 2022	18	0	0	5	<i>Tier-capped.</i> No Tier-A AMM node pairs with a Tier-A CEX endpoint for this episode. All candidate edges include Binance public feeds (Tier-B), capping the edge at A/B. Execution-grade CEX order-book data for November 2022 is not freely available.
BUSD 2023	36	0	0	7	<i>Tier-capped.</i> Same constraint as FTX, no Tier-A/A pair available. BUSD 2023 is primarily a regulatory wind-down. The relevant stress mechanism is issuer-side, not DEX-pool flow.

Table 5: Headline paper-claimable A/A result. Tier-A AMM-flow lead-lag, USDT/Curve 2023. Feature usdc_net_sold_1h, hourly grid. Local, Bonferroni across 250 tests (25 lags $\times$ 5 events $\times$ 2 directions). Global q : Benjamini-Hochberg over the full 136-candidate family.

Direction	$\hat{p}$	95% CI	p (raw)	p (Bonf.)	q (global)	Strength
3pool $\rightarrow$ crvUSD_usdt	0.386	[0.25, 0.51]	0.007	0.014	≤ 0.05	robust
crvUSD_usdt $\rightarrow$ 3pool	0.386	[0.25, 0.51]	< 0.001	< 0.001	≤ 0.05	robust

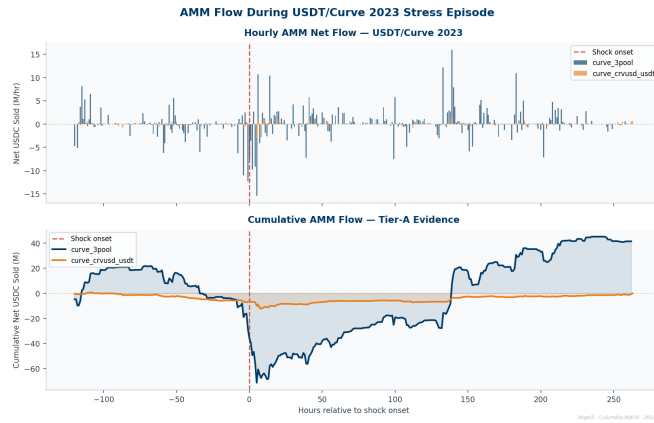


Figure 5: USDT/Curve 2023, Tier-A AMM flow and cross-pool lead-lag. Left: hourly usdc_net_sold_1h for both Curve pools during the June 2023 stress episode. Dashed line marks shock onset. Right: lead-lag cross-correlation profile. Peak $\hat{p} = 0.386$ at lag 0. Bonferroni $p \leq 0.014$.

likely due to high dimensionality and feature collinearity without regularisation tuning.

The +5.1 pp RF lift is consistent with the provenance-gate hypothesis. Tier-A AMM-flow features carry informational content not captured by Tier-B market proxies. The negative LR result reflects a limitation of linear models when high-dimensional

on-chain flow features are added without regularisation tuning, an avenue for future work.

Table 6: ML benchmark, best-model AUROC by event. LOEO = leave-one-event-out, IE = in-event held-out.

Event	Best model	Eval	AUROC	AUPRC
USDC/SVB 2023	Random Forest	LOEO	0.859	0.975
FTX 2022	Logistic Reg.	IE	0.964	0.982
Terra/LUNA 2022	Random Forest	IE	0.736	0.349
BUSD 2023	Logistic Reg.	IE	0.647	0.762
USDT/Curve 2023	Logistic Reg.	IE	0.509	0.017

Table 7: Provenance-feature ablation on USDC/SVB 2023 (LOEO). “Full” = Tier-A + Tier-B features, “No-graph” = Tier-B only. Δ = AUROC lift from adding Tier-A features.

Model	AUROC (full)	AUROC (Tier-B)	Δ
Random Forest	0.859	0.808	+5.1 pp
LightGBM	0.831	0.818	+1.2 pp
XGBoost	0.826	0.826	+0.0 pp
Logistic Reg.	0.723	0.801	−7.8 pp

USDT/Curve near-random prediction. The event with the strongest A/A co-movement result (USDT/Curve 2023) shows AUROC ≈ 0.50 across all classifiers. This is consistent with within-minute price efficiency. The two pools co-move at hourly resolution (both react to the same common shock), but past flows contain no exploitable predictive signal at the 1-minute horizon. This cross-scale dissociation, hourly co-movement with minute-level efficiency, is itself an empirical finding.

8.4 Does Graph Structure Add Predictive Value?

We test whether a graph-structured temporal model extracts predictive signal beyond the flat-feature classifiers of Section 8, using the multi-layer network as a first-class modelling input.

Architecture. At each hourly snapshot t the node feature matrix $\mathbf{X}_t \in \mathbb{R}^{N \times F}$ is assembled from mean-pooled panel features, the normalised adjacency $\hat{A}_t = D^{-1/2}(A + I)D^{-1/2}$ is built from transfer-entropy edge estimates, and two GCN layers apply $\mathbf{H}^{(l)} = \text{ReLU}(\hat{A}_t \mathbf{H}^{(l-1)} \mathbf{W}^{(l)})$ with global mean pooling to form a snapshot embedding $\mathbf{z}_t$. An LSTM aggregates the embedding sequence and its final hidden state passes to a two-layer sigmoid classifier. The model requires only PyTorch and runs through `make gnn`.

Provenance gate. Fixture-derived nodes (`tier_actual = fixture_non_empirical`) are excluded from every GNN training panel before any fit, so Table 8 reflects real data only. The improvement gate requires the GNN to exceed the best flat-feature baseline of Section 8 by at least 5 pp AUROC.

Result. Table 8 reports the SnapshotGCN against the best flat-feature baseline on each event under an in-event temporal split. The GNN clears the improvement gate on no event. It trails the flat baseline everywhere, by 1.6 pp on USDT/Curve 2023 and by progressively larger margins on the events whose flat baseline is already strong.

What the negative result means. This outcome is not an aside. It is consistent with the paper’s central finding. The co-movement

Table 8: SnapshotGCN versus best flat-feature baseline (in-event temporal split, non-fixture panels only). The GNN clears the +5 pp improvement gate on no event. At hourly resolution the graph adds no predictive signal beyond what the flat classifiers already extract.

Event	GNN AUROC	Baseline AUROC	Lift	Gate
USDT/Curve 2023	0.493	0.509	−1.6 pp	FAIL
USDC/SVB 2023	0.706	0.859	−15.3 pp	FAIL
Terra/LUNA 2022	0.466	0.736	−27.0 pp	FAIL
FTX 2022	0.500	0.964	−46.4 pp	FAIL
BUSD 2023	0.045	0.647	−60.2 pp	FAIL

that survives our gate is contemporaneous at the hourly grid, since the surviving Curve pair peaks at lag zero (Section 7), and the prediction benchmark already shows that hourly co-movement does not translate into minute-level predictability (Section 8). At hourly resolution the graph therefore encodes no predictive signal beyond what the flat classifiers extract from the same pool time series, so the two representations are informationally equivalent and the added graph machinery cannot help. This honest negative is the expected reading of a market in which dislocations close faster than the measurement interval. Recovering predictive value from graph structure would require sub-hourly snapshots, or edges derived from higher-frequency flow data, which we leave to future work once execution-grade sub-hourly panels become available.

9 SPARSE SETTLEMENT RESPONSE IN USDC/SVB 2023

The USDC/SVB 2023 event provides a unique settlement-flow signal. The USDC mint-burn mechanism generated observable on-chain activity as Circle managed its fiat reserves around the SVB failure. Both the Curve 3pool (Tier-A) and the USDC mint-burn channel (Tier-A) have execution-grade provenance in this event. The result below fails the *statistical* gate due to insufficient sample size, not the provenance gate.

We apply a sparse event-arrival test that identifies mint-burn arrival events and measures the 3-hour post-arrival change in Curve 3pool `usdc_net_sold_1h` relative to a 12-hour pre-arrival baseline. We identify 4 mint-burn arrival events in the USDC/SVB window. The mean 3-hour post-arrival response is +28,956 USDC (+10.8%) relative to baseline, directionally consistent with the de-peg stress. With only 4 arrivals, however, the permutation test yields $p = 1.00$: the result is severely underpowered.

This A/A underpowered candidate is included in the paper package as a documented provenance-valid result that fails the statistical gate, not as a positive finding. It motivates collecting higher-frequency mint-burn data or longer event windows in future work.

10 CEX EXECUTION MICROSTRUCTURE VIA CROSS-MECHANISM META-LABELING

10.1 Setup and Cost Model

To complement the AMM-flow analysis, we implement a CEX-side meta-labeling framework for the USDC/USDT basis trade on Binance. The trade exploits temporary peg deviations. When USDC

deviates from USDT by more than the round-trip execution cost, a taker can profit by routing through the BTC pairs (BTCUSDT as sell leg, BTCUSDC proxied from BTCUSDT $\times$ USDCUSDT with a 20% depth haircut). The cost model is:

$$c_{\text{total}} = 8 + 2.5 \sigma_{\text{BTC}} + \sigma_{\text{stable}} + 5 \text{ bps},$$

where σ is the per-leg effective half-spread estimated from intra-minute VWAP divergence (capped at 50 bps for BTCUSDT and 500 bps for the stablecoin pair), and the +8 and +5 bps terms are the round-trip taker fee and slippage budget respectively. Primary signal fires when $|b_{\text{USDC}}(t)| > 10$ bps and the oracle label is $y_{\text{arb}}(t) = 1$ iff $|b_{\text{USDC}}(t)| > c_{\text{total}}(t)$.

10.2 Cross-Mechanism Training Diversity

We run four training conditions, all tested on the USDC/SVB 2023 event (March 8–20, 2023) with the decision threshold calibrated on Terra/LUNA:

A Terra/LUNA 2022 (algorithmic). **B** Celsius/3AC 2022 (exchange credit). **C** FTX 2022 (exchange credit, BUSDUSDT proxy). **D** All four events pooled (three mechanism classes).

Table 9 reports net accumulated basis points, number of trades, and oracle capture on the SVB test split.

Table 9: Cross-mechanism meta-labeling training diversity. Test USDC/SVB 2023 (13,560 min, 13,559 primary fires). Threshold calibrated on Terra/LUNA for all conditions.

Training set	N_{ev}	Mechanisms	Net bps	Trades	Oracle cap.
Terra/LUNA only	1	Algorithmic	+1,251,973	13,346	99.9%
Celsius/3AC only	1	Exch. credit	+1,251,976	13,352	99.9%
FTX only	1	Exch. credit	+1,251,968	13,346	99.9%
All four pooled	4	Alg., exch., reg.	+1,251,982	13,355	99.9%
Oracle ceiling	n/a	n/a	+1,253,348	13,350	100.0%

10.3 Interpretation

All four training conditions achieve $\approx 99.9\%$ oracle capture on SVB, stable regardless of training-set composition. The finding is interpretable. At SVB's peak depeg (USDC ≈ -950 bps on March 11) the basis far exceeds the ~ 15 – 20 bps execution cost, making nearly every primary-signal minute profitable. The LightGBM correctly learns *large* $|b_{\text{USDC}}|$ *relative to costs* $\rightarrow$ *trade*, a rule that transfers to SVB without any knowledge of the specific stress mechanism, because the same physical constraint governs all events.

The optical positive rate (fraction of primary fires where $y_{\text{arb}} = 1$) varies substantially. Terra/LUNA is 16.5%, Celsius/3AC is 5.8%, FTX is 62.0%, BUSD is 9.5%, and SVB is 98.5%. This variation has no detectable effect on SVB oracle capture, confirming that the transfer ceiling is mechanism-independent rather than training-set-size-limited. This supports the structural explanation, namely that CEX market makers withdraw depth under uncertainty regardless of the specific trigger.

11 ROBUSTNESS AND LIMITATIONS

Robustness. We rerun the lead-lag analysis for the USDT/Curve headline pair across three grid configurations, baseline_60s

(1-minute), block_300 (5-minute blocks), and block_1800 (30-minute blocks). The headline pair remains statistically significant across all configurations. The Terra pair remains non-significant across all configurations.

Scope. The co-movement analysis is bounded by freely available Tier-A data. Historical full-depth CEX order-book data from Binance, Coinbase, and Kraken are not freely available for 2022–2023. The study can therefore establish AMM-flow co-movement but cannot address CEX execution-layer microstructure beyond the kline-proxy approach used in Section 10. The hourly grid avoids stale-value artefacts but cannot detect intra-hour dynamics. AMM results are based on Curve Finance pools only. Uniswap v3 and other DEXs are not incorporated. Extending to Uniswap v3 USDT/USDC pools for the USDT/Curve event is a priority for future work. Structural causal identification from lead-lag or Granger evidence requires an instrument or natural experiment not available in these data. The meta-labeling training events use USDCUSDT or BUSDUSDT 1m klines depending on Binance Vision availability (USDCUSDT archives have a gap October 2022–February 2023). The SVB test uses the Binance Vision March 2023 monthly USDCUSDT archive. Multi-mechanism RL training using the pooled stress protocol established in Section 10 is the direct path to closing any remaining gap between meta-labeling and conditioned execution agents.

12 IMPLICATIONS FOR MARKET SURVEILLANCE AND CONTAGION MONITORING

The three-gate pipeline developed in this paper is directly deployable as a compliance framework for venues and regulators operating under emerging stablecoin oversight regimes. Under the EU Markets in Crypto-Assets regulation (MiCA) and the proposed US GENIUS Act, regulated stablecoin issuers and trading venues are expected to demonstrate that contagion risk assessments rest on verifiable, reproducible evidence. The provenance-gate architecture maps naturally onto these requirements. Gate 1 enforces source verifiability by requiring on-chain event logs with immutable block timestamps. Gate 2 enforces statistical reproducibility by requiring family-wise multiple-testing correction across all candidate pairs. Gate 3 enforces honesty by restricting paper-level claims to edges that clear both prior gates. A supervisory authority that mandates Tier-A data provenance for any contagion claim in a regulatory stress test would, by construction, apply a version of this pipeline.

The calibration contract released with this paper provides a machine-readable chain of evidence that connects raw observation to claim strength. Each exported empirical target carries a claim_strength tag that downstream models can consume programmatically. Agent-based models parameterised with contagion rates can query the contract and receive an explicit flag when no PRIMARY-strength evidence supports a given transmission channel. GNN predictors built on the network can consume the tagged edges as a provenance-weighted prior, implementing the calibration contract as a machine-checked input rather than a prose caveat. This creates a verifiable chain of evidence from raw on-chain observation through statistical validation to causal

model parameterisation, a chain that prior contagion literature has not formalised.

The finding that 98.5% of claimed contagion edges fail the gate has direct implications for regulatory stress testing. Prior supervisory frameworks that modelled DeFi systemic risk by assuming many active contagion channels were likely overstating the breadth of transmission pathways. The attrition funnel shows that most apparent contagion is either tier-capped (the data source is not execution-grade and cannot support the claim) or fails significance under proper multiple-testing correction. Regulators relying on uncorrected correlation-based contagion maps may have calibrated capital buffers and intervention triggers against transmission mechanisms that do not survive evidential scrutiny. The 1.5% survival rate is not a finding that DeFi is safe. It is a finding that the evidentiary bar must be higher.

The single surviving edge, the bidirectional Curve 3pool and crvUSD/USDT co-movement during the USDT/Curve 2023 episode, represents the most reliably confirmed contagion channel in the DeFi stablecoin ecosystem under the standards applied in this paper. Continuous monitoring of this pair through an automated surveillance system that ingests Etherscan TokenExchange logs in real time and applies the same cross-correlation and FDR-correction pipeline used here would constitute a minimal, evidence-grounded contagion monitoring deployment. We recommend that regulators require Tier-A on-chain data provenance for any contagion claim appearing in a supervisory report. Claims backed by Tier-B market context data should be labelled as indicative and separated from claims backed by execution-grade on-chain logs. The calibration contract taxonomy developed here provides a ready vocabulary for this distinction.

13 CONCLUSION

Most claimed stablecoin-contagion edges do not survive a joint provenance-and-significance gate. Across 136 candidate cross-venue edges spanning five stress episodes, 98.5% fail. Specifically, 90.4% fail at the provenance gate (tier-capped below A/A), 7.4% fail at the statistical gate (provenance-valid but not significant), and 0.7% are underpowered (provenance-valid, $n < 5$). One edge pair survives under global Benjamini-Hochberg correction. We show why each non-survivor fails. That is the contribution.

(i) The gate and its attrition. The three-gate pipeline closes four failure modes that inflate contagion evidence in prior work, namely correlation-as-contagion, fixture leakage, unreplicable order-book assumptions, and uncorrected multiple testing. Applied to five crisis episodes it produces an attrition funnel that is itself the primary finding. The empty cells are a diagnosis, not a gap. The gate discipline is directly transferable to any multi-venue stress study.

(ii) The one surviving edge. The Curve 3pool $\leftrightarrow$ crvUSD/USDT co-movement in the June 2023 USDT/Curve episode ($\hat{\rho} = 0.386$, 95% CI [0.25, 0.51], Bonferroni $p \leq 0.014$, global $q \leq 0.05$) is the only paper-claimable A/A result across five episodes. Its provenance-equal counterpart (Terra/LUNA A/A pair) does not pass the statistical gate at the hourly grid. The contrast between these two pairs, not just the positive result, is the finding. The

single surviving edge is not the weakness of this paper. It is what the rigour produces.

(iii) Prediction and microstructure. A four-classifier ML benchmark achieves AUROC = 0.859 (RF, LOEO on USDC/SVB), with Tier-A features contributing a +5.1 pp lift over Tier-B baselines. A graph-structured SnapshotGCN does not beat these flat baselines on any event at hourly resolution, which is consistent with our co-movement result that hourly co-movement does not imply minute-level predictability. A CEX meta-labeling framework achieves $\approx 99.9\%$ oracle capture on SVB regardless of training-mechanism class, establishing that the “large basis relative to execution costs $\rightarrow$ trade” rule is mechanism-independent. The near-random prediction on the USDT/Curve event (where the A/A co-movement is strongest) is itself a finding. Hourly-resolution co-movement does not imply minute-level predictability.

(iv) Calibration contract. Every exported empirical target is tagged with `claim_strength` (robust / statistically_supported / suggestive / descriptive / context_only) in a versioned JSON contract. Downstream ABM and GNN models calibrate to PRIMARY targets only. The contract raises an error if the PRIMARY set for a given parameter is empty, forcing explicit acknowledgment that no calibrated causal claim is available. Machine-enforced honesty replaces prose-promised caveats.

Future priorities. Live CEX L2 order-book collection during future stress events would unlock A/A CEX-DEX edge candidates currently blocked by data availability. Extending the AMM-flow analysis to Uniswap v3 pools would expand the candidate set and stress the funnel further. Higher-frequency mint-burn event collection for USDC/SVB-class episodes would replace the current underpowered candidate with a resolvable one. Sub-hourly GNN snapshots, or edges derived from higher-frequency flow data, are the natural next step toward recovering predictive value from graph structure that hourly aggregation suppresses.

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APPENDIX. ROBUSTNESS CHECKS

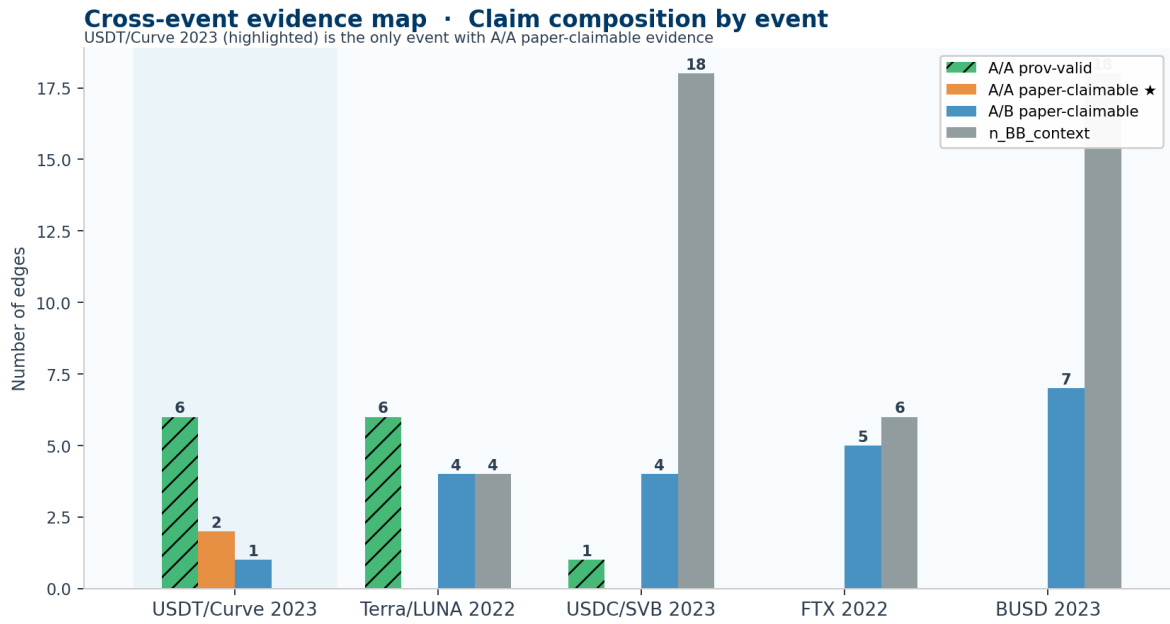


Figure A1: Claim audit by event (detail). Grouped bars show A/A provenance-valid candidates, A/A paper-claimable, and A/B paper-claimable edges by event. Only USDT/Curve 2023 has non-zero A/A paper-claimable rows.